

Lakshyaraj Singh Rao

Backend Engineer / AI Infrastructure / Full-Stack / DevOps

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SUMMARY

I build the reliability layer for production AI. Retrieval that **refuses** when it has no source. Streams that **abort** when output goes off-script. Evals that **gate merges**. Webhooks that stay **idempotent**. Co-Founder at **Homesty.ai LLP** since Nov 2025 on Next.js 15 + pgvector + GPT-4o. Four **@ykstorm** npm packages (two with SLSA provenance) and a GitHub Marketplace Action, each pulled from a specific production failure.

EXPERIENCE

Homesty.ai LLP

Nov 2025 to Present

Co-Founder & Founding Engineer · Mumbai, India

Built the buyer-facing AI chat end-to-end on **Next.js 15**, **Postgres + pgvector**, **Prisma 7**, **GPT-4o** and **Claude streaming**. Wrote the retrieval layer that refuses without a grounded source and the token-level guard that aborts mid-stream. Own Sentry, on-call, and the audit log pipeline. Anchor and Tripwire were pulled out of this codebase as npm packages that now run back inside it.

PROJECTS

Anchor / provenance-first RAG library that refuses to hallucinate

anchor-iota-ten.vercel.app/playground · github.com/ykstorm/anchor

Every RAG demo breaks the moment retrieval finds nothing useful and the model makes things up. Anchor returns chunks with provenance above the similarity floor, and `refused: true` below. Built inside a real-estate app where a hallucination is a lawsuit. **p50 2.6 ms retrieval on 100k vectors (HNSW), measured in CI.**

Anvil / idempotent webhook engine with dedup, retry, and dead-letter

github.com/ykstorm/anvil · npmjs.com/package/@ykstorm/anvil

Stripe re-delivers the same webhook three times on flaky networks and your worker double-charges the user. Anvil sits in front of every webhook source with dedup, retry, and a dead-letter queue, so the handler runs exactly once. Shipped as **@ykstorm/anvil v0.1.1** with **SLSA provenance**, plus a Terraform module and Helm chart. **10.5k req/s sustained ingress, constant-time HMAC verify at 0.9% timing delta, measured in CI.**

Goldset / CI for AI evaluation, published on GitHub Marketplace

github.com/ykstorm/goldset · github.com/marketplace/actions/goldset

The usual answer to "did my prompt change break anything?" is "let me eyeball the notebook." Goldset runs three eval runners on every PR (**golden answers, LLM-as-judge, structural assertions**), posts a delta-vs-base comment, and blocks merge on regression. Shipped as **@ykstorm/goldset v0.2.4** on npm and GitHub Marketplace. **Deterministic runners clear 1k cases in 3 ms, so the gate never slows a PR.**

Tripwire / mid-stream LLM guardrail that kills bad tokens before they render

github.com/ykstorm/tripwire · npmjs.com/package/@ykstorm/tripwire

Once a bad token reaches the user, post-hoc moderation is too late. Tripwire watches every token mid-stream and aborts on rule trip: PII leaks, prompt injection, fabricated facts. Drop in as an OpenAI-compatible proxy (one URL change) or import as a library. Shipped as **@ykstorm/tripwire v1.1.0** on npm plus Docker at ghcr.io/ykstorm/tripwire. **4.7 µs per chunk overhead, zero buffering on the happy path, measured in CI.**

Stackup / production Kubernetes stack on a laptop, free, in under 15 minutes

github.com/ykstorm/stackup · ykstorm.github.io/stackup

Managed Kubernetes starts at \$200 a month. Stackup runs the same loop teams pay for on your laptop: **kind + ArgoCD app-of-apps + Argo Rollouts + Prometheus/Grafana**, plus Sealed Secrets, cert-manager, and Pod Security Standards `restricted`. One `make up`, six child apps synced by GitOps, canary shifts 25 → 50 → 75 → 100 with automated rollback. **8/8 canary steps healthy under real Prometheus success-rate gate at ≥0.95, verified in CI.**

TECHNICAL SKILLS

Languages TypeScript, JavaScript, Python, SQL, Bash

Backend & AI Node.js, **Next.js 15**, Express, **Postgres + pgvector**, Prisma 7, Redis, BullMQ, SSE / WebSockets, **OpenAI**, Anthropic, Ollama, RAG, mid-stream guardrails, LLM eval, SLSA provenance

Cloud, K8s & CI/CD **Docker**, **Kubernetes**, Helm, kind, **ArgoCD**, Argo Rollouts, Sealed Secrets, cert-manager, Terraform (AWS + Hetzner), GitHub Actions, Vercel

Observability & Security **Prometheus**, Grafana, Sentry, HMAC (constant-time), rate limiting, prompt-injection defense, NetworkPolicy, Pod Security Standards

EDUCATION

Manipal University Jaipur · B.Tech, Computer Science

Jaipur, India · Graduating 2026

Relevant coursework: Operating Systems, Computer Networks, Database Management Systems, Object-Oriented Programming, Cloud Computing